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Impact of Anti-Fatigue Floor Mat on Surgical Staff Comfort Levels in Head and Neck Surgery Cases

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ABSTRACT

Objective: Musculoskeletal symptoms are common among surgical staff and can have long-term implications on health and wellbeing. The purpose of this study is to evaluate the impact of anti-fatigue floor mat on the comfort level of surgical teams during head and neck surgeries lasting ≥ 3 h.

Methods: Over 4 months, we prospectively randomized 34 major (≥ 3 h) head and neck procedures to the use or not of an anti-fatigue floor mat. Anonymous questionnaires measured the comfort levels in different subjects including the surgeons, assistant surgeons, and surgical scrub technicians ($n = 57$). Subjects completed questionnaires before, immediately after, and one day after surgery. Variables collected included demographics, overall discomfort level, overall energy level, discomfort level in different body parts, number of breaks taken during the case, time since last physical exercise, and frequency of physical exercise. The analysis of variance (ANOVA) technique was used for data analysis.

Results: The group that used anti-fatigue floor mats reported lower increases in discomfort from pre-op to immediately post-op and 24h post-op compared to the group that did not ($p = 0.009$ and $p < 0.001$). Participants who used the mats reported significant lower levels of pain in the ankles and feet, knees, and shoulders immediately post-op compared to participants who did not. Participants who used the mats reported lower increases in discomfort in their back, hips, knees, neck, and shoulders from pre-op to post-op.

Conclusions: Using anti-fatigue floor mats during surgery is an effective and low-cost intervention to decrease the musculoskeletal symptoms experienced by the surgical team.

Level of Evidence: 2.

1 | Introduction

Awareness surrounding healthcare workers' health and wellness has significantly increased in recent years. Work-related fatigue and body strain is common among surgeons and operating room staff and can have long-term implications on a person's

health and wellbeing. Over time, this strain can lead to time off work, medical and/or surgical intervention, and even early retirement [1].

Musculoskeletal fatigue is common among surgeons and operating room staff due to the considerable lengths of time spent

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standing and working in static positions [2]. Previous studies have found that over 90% of otolaryngology residents experience at least one musculoskeletal symptom at any point in time, ranging from upper body pain (including shoulder, neck, and back pain) to lower body pain (hip, thigh, and foot pain) [3]. In one study of 141 otolaryngologists, 6.4% of residents reported missing work due to musculoskeletal symptoms, and 16.3% of residents reported stopping during an operation due to musculoskeletal pain. 88.3% of these residents attributed these musculoskeletal symptoms to their surgical training [4]. Other studies found that approximately 95% of the surgeons surveyed had experienced at least one work-related musculoskeletal symptom within the last six months [5, 6]. The likelihood of experiencing pain and fatigue increases as surgery length increases [3].

Historically, healthcare workers have some of the highest rates of occupational injuries. Nationwide, hospitals pay \$0.78 in workers compensation for every \$100 in payroll, resulting in approximately \$2 billion annually in workers' compensation for occupational injuries [7]. Reducing the burden of musculoskeletal fatigue provides an opportunity for hospitals to potentially save billions of dollars each year.

Previous studies have suggested that the use of anti-fatigue floor mats in the operating room can improve energy levels and decrease discomfort for surgeons both during surgery and postoperatively [5]. This study evaluates the effectiveness of anti-fatigue floor mats at reducing the surgical team fatigue and discomfort during head and neck operations lasting ≥ 3 h. The incorporation of anti-fatigue floor mats into operating rooms could decrease both short-term and long-term physician fatigue and discomfort, leading to improved surgical performance and improved overall wellbeing.

2 | Materials and Methods

This study was a prospective randomized controlled trial of head and neck surgical cases lasting ≥ 3 h. Subjects were limited to head and neck surgeons, assistant surgeons (residents/fellows), and surgical scrub technicians within the Otolaryngology department at the University of Oklahoma Health Sciences Center from August 2021 to November 2021. The study protocol was approved by the University of Oklahoma Health Sciences Center Institutional Review Board.

This was a two-arm study: During the first arm, no surgical mats were used in the operating room. During the second arm, cases which were allotted a time ≥ 3 h on the surgery schedule were randomly selected to have anti-fatigue floor mats in the operating room. The otorhinolaryngology surgery schedule was obtained on a weekly basis from the surgical scheduler on staff. Only the cases which did surpass the 3-h criteria were ultimately included in data analysis.

The anti-fatigue floor mats that were used for the study were medical grade mats from the brand name GelPro. Mats used were 20" $\times$ 32" in size. Retail price was 167 USD with tax per mat.

The surgical team members involved in the case completed an anonymous questionnaire (Data S1) at three different points in

time: pre-operatively, immediately post-operatively, and 24 h postoperatively. Informed consent was obtained from participants by means of approval to complete the questionnaire. Every participant was given three copies of the questionnaire for each case: one for pre-operation, one for post-operation, and one for 24 h post-operation. The study investigator kept a log which included the questionnaire number, the date of the procedure, the operating room number, and the presence of anti-fatigue mat or not. The participants returned the completed questionnaires to a mailbox kept in a secure location in the operating room.

The variables collected on this questionnaire included: date of procedure, status at the time of the survey completion (pre-op/post-op/24 h post-op), position on the surgical team (attending, resident, scrub tech), age (in years), surgical experience (in years), duration since last day of exercise (days), frequency of exercise regimen (number of days per week), case number of the day (first, second, etc), number of breaks taken during case, presence of anti-fatigue mats, overall discomfort level, overall level of energy, and level of discomfort/pain in different body parts (feet, ankle, knees, hips, back, shoulders, neck) (Figure 1).

Background characteristics of the surgical team participants were summarized using counts and proportions for categorical and mean and standard deviation for continuous variables.

Item on Questionnaire	Units
Date of procedure	
Status at the time of survey completion	pre-op/ post-op/ 24 hours post-op
Position on the surgical team	attending/ resident/ scrub technician
Age	years
Surgical experience	years
Duration since last day of exercise	days
Frequency of exercise regimen	days per week
Case number of the day	first/ second/ third
Number of breaks taken during the case	1-3
Presence of anti-fatigue floor mat	yes/no
Overall discomfort level	1-10
Overall energy level	1-10
Level of discomfort/pain in feet	1-10
Level of discomfort/pain in ankles	1-10
Level of discomfort/pain in knees	1-10
Level of discomfort/pain in hips	1-10
Level of discomfort/pain in back	1-10
Level of discomfort/pain in shoulders	1-10
Level of discomfort/pain in neck	1-10

FIGURE 1 | Questionnaire values.

Similarly, the mean and standard deviation of all 8 parameters of fatigue measurements were estimated for baseline, immediately after surgery, and 24 h after surgery. The difference in the fatigue scores (immediately after surgery-baseline, 24 h after surgery-baseline, and 24 h surgery-immediately after surgery) according to the use of an anti-fatigue gel mat (yes/no) were estimated and the association between change in fatigue scores and gel mat use was assessed using the analysis of variance (ANOVA) method. Analyzed values sets had a normal distribution. Measured observations were independent since the groups were randomized. Statistical significance was set at $p < 0.05$ and the statistical analysis was performed using SAS 9.4.

3 | Results

Assuming a total 20% difference in the mean overall discomfort level between the two groups, an alpha of 0.05, and power of 0.8, it was determined that at least 32 procedures in total, with 16 procedures in each group, were required for data analysis.

Analysis of the results found that the anti-fatigue floor mats were effective at reducing the discomfort felt by the surgical team members both immediately post-operation and 24 h post-operation when compared with the surgical team members who did not use the floor mats during surgery.

These results reflect survey responses across 34 operations with an $n = 57$. Analysis of the study population characteristics revealed no significant differences in age ($p = 0.587$), duration since exercise ($p = 0.600$), years of operating room experience ($p = 0.209$), or number of breaks taken ($p = 0.364$) between the participants who used the mats in the operating room and those who did not (Figure 2). There was a slight significance found between the two groups for frequency since exercise, with the group using the mats having an average frequency of 2.5 ± 1.8 and the group not using the mats having an average frequency of 2.0 ± 0.8 ($p = 0.048$).

When comparing the baseline survey responses between the two groups pre-operatively, there were no significant differences

Variable		Without Mats	With Mats	Parametric p-value
Age	Mean	38 ± 11	37 ± 10	0.587
	Median	33	34	
Duration Since Exercise	Mean	3.4 ± 3.8	3.1 ± 2.9	0.600
	Median	3	2	
Frequency of Exercise	Mean	2.5 ± 1.8	2.0 ± 0.8	0.048
	Median	2	2	
Years OR Experience	Mean	9.4 ± 6.9	8.0 ± 4.7	0.209
	Median	8	8	
Number of Breaks Taken	Mean	0.5 ± 0.6	0.3 ± 0.6	0.364
	Median	0	0	

FIGURE 2 | Characteristics of the study population.

in the overall energy ($p = 0.650$), overall discomfort ($p = 0.392$), ankles/ft discomfort ($p = 0.981$), knee discomfort ($p = 0.087$), hip discomfort ($p = 0.935$), back discomfort ($p = 0.497$), shoulder discomfort ($p = 0.734$), or neck discomfort ($p = 0.663$) between the participants who used the mats and those who did not use the mats (Figure 3).

In every category, the mean discomfort levels immediately post-operation was lower for the participants using the mats compared to the participants not using the mats. This trend remains true even for the values that did not reach statistical significance. The levels of ankle/ft discomfort ($p = 0.012$), knee discomfort ($p = 0.005$), and shoulder discomfort ($p = 0.044$) were significantly lower for the participants who used the anti-fatigue floor mats during surgery compared to those who did not use the floor mats during surgery (Figure 4).

Comparing the change in discomfort levels from pre-operation to immediately post-operation revealed that the participants who used the anti-fatigue floor mats experienced significantly smaller increases in their overall discomfort ($p = 0.009$), ankles/ft discomfort ($p = 0.004$), knee discomfort ($p = 0.035$), hip discomfort ($p = 0.047$), back discomfort ($p = 0.012$), shoulder discomfort ($p = 0.005$), and neck discomfort ($p = 0.017$)

Variable	Without Mats (n=30)	With Mats (n=27)	Parametric p-value
Overall Energy	7.4	7.6	0.650
Overall Discomfort	2.0	2.3	0.392
Ankles/Feet Discomfort	1.6	1.6	0.981
Knee Discomfort	2.1	1.4	0.087
Hip Discomfort	1.7	1.7	0.935
Back Discomfort	1.8	2.0	0.497
Shoulder Discomfort	1.9	2.0	0.734
Neck Discomfort	1.7	1.9	0.663

FIGURE 3 | Baseline pre-operation energy and discomfort levels.

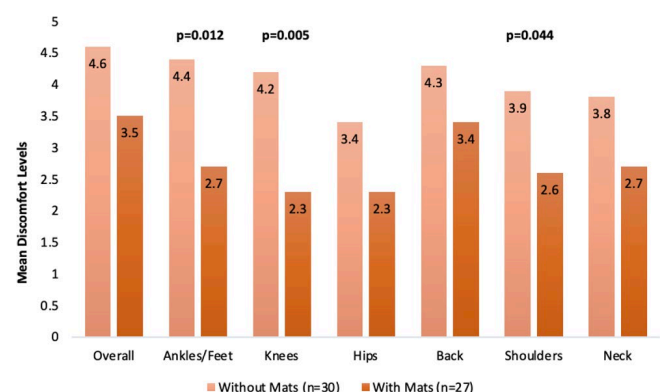


FIGURE 4 | Mean discomfort levels immediately post-operation.

Variable	Δ Without Mats (n=30)	Δ With Mats (n=27)	Parametric p-value
Overall Energy	-2	-1.5	0.193
Overall Discomfort	2.6	1.2	0.009
Ankles/Feet Discomfort	2.8	1.2	0.004
Knee Discomfort	2.0	0.9	0.035
Hip Discomfort	1.6	0.6	0.047
Back Discomfort	2.5	1.3	0.012
Shoulder Discomfort	2.0	0.6	0.005
Neck Discomfort	2.1	0.8	0.017

FIGURE 5 | Change in discomfort levels from pre-op to immediately post-op (mean values).

Variable	Δ Without Mats (n=30)	Δ With Mats (n=27)	Parametric p-value
Overall Energy	-1.7	1.0	0.040
Overall Discomfort	1.9	-1.0	<0.001
Ankles/Feet Discomfort	1.3	0.5	0.193
Knee Discomfort	1.0	0.3	0.233
Hip Discomfort	0.4	-0.5	0.078
Back Discomfort	1.3	-1.0	0.011
Shoulder Discomfort	0.9	-1.0	0.028
Neck Discomfort	0.9	0.3	0.166

FIGURE 6 | Change in energy and discomfort levels from pre-op to 24 h post-op (mean values).

compared to the participants who did not use the floor mats in the operating room (Figure 5). It stands to reason that any member of the surgical team will feel more uncomfortable and more fatigued after surgery than they did before surgery. These response values suggest use of the anti-fatigue floor mats mitigated the severity of discomfort and energy less felt during surgery.

The change in overall energy levels from pre-operation to 24 h post-operation was significantly greater for the participants who used the mats during surgery compared to those who did not use the mats ($p=0.040$). The increase in discomfort levels from pre-operation to 24 h post-operation were significantly lower for the participants who used the mats compared to those who did not when evaluating overall discomfort ($p<0.001$), back discomfort ($p=0.011$), and shoulder discomfort ($p=0.028$) (Figure 6).

Multivariate analysis revealed that back, hip, and knee discomfort are significantly correlated with age, exercise regimen, years of operating room experience, and the number of breaks taken during the case (Figure 7). Change in back discomfort from pre-operation to post-operation was statistically

significant when analyzed alongside age ($p=0.004$), duration since last exercise ($p=0.004$), frequency of exercise ($p=0.004$), years of OR experience ($p=0.007$), and number of breaks taken ($p=0.028$). Similarly, change in hip discomfort from pre-operation to post-operation was statistically significant when analyzed alongside age ($p=0.003$), duration since last exercise ($p=0.003$), frequency of exercise ($p=0.003$), years of OR experience ($p=0.004$), and number of breaks taken ($p=0.028$). Change in knee discomfort from pre-operation to post-operation was statistically significant when analyzed alongside years of OR experience ($p=0.049$) and number of breaks taken ($p=0.011$). These results indicate a need for further study into mitigating back, hip, and knee discomfort, particularly among older surgeons.

4 | Discussion

The use of anti-fatigue floor mats in the operating room was found to be effective in reducing the discomfort levels felt by members of the surgical team immediately post-operation and 24 h post-operation. The surgical team members who used the anti-fatigue floor mats in the operating room reported significantly lower levels of discomfort in their ankles, feet, knees, and shoulders immediately post-operatively compared to those who did not use the floor mats. It is useful to note that the discomfort levels in every part of the body were lower for the participants who used the mats, even those that did not reach statistical significance. Participants who used the floor mats during surgery experienced smaller increases in the discomfort between pre-operation and post-operation in the following body regions: ankles, feet, knees, hips, back, shoulders, neck, and overall body.

Previous studies have identified similar results using anti-fatigue floor mats in various occupational settings. A 2002 study of assembly line workers, who also stand in one static position during long workdays, found statistically significant correlations between the type of flooring used and pain/discomfort level when comparing hard floor and mats. The flooring types included shoe insoles, floor mats, hard block floors, and a shoe insole plus floor mat combination. Participants reported the highest levels of discomfort on the hard block floors, and the lowest levels of discomfort using the shoe insole plus floor mat combination [8]. A 2004 study of factory workers found that standing on floor mats produced less discomfort compared to standing on a hard wooden floor. The use of shoe insoles was found to be the most effective intervention for reducing discomfort, followed second by the use of anti-fatigue floor mats [9]. A 2016 study of surgeons using different flooring options in the operating room found that 70% of the surgeons said they would recommend the use of anti-fatigue mats to a friend, 65% of surgeons preferred the use of a floor mat to hard concrete floors, and 45% said the mat helped reduce their musculoskeletal related symptoms [10].

There are numerous potentially confounding factors that could contribute to the level of fatigue experienced by the surgical team [11]. One potential confounding factor is the choice of footwear among the surgical team members. This is highlighted in the previous studies mentioned in which the use of shoe insoles was found to significantly reduce discomfort. Another potentially confounding factor could be the amount of rest received by

discomfort_Back_1_0					
Variable	N	Pearson CC	Pearson P-value	Spearman CC	Spearman P-value
Age	55	0.378	0.004	0.381	0.004
Duration_since_last_day_of_exerc	55	-0.091	0.004	0.006	0.004
Frequency_of_exercising__days_we	55	-0.193	0.004	-0.089	0.004
Years_of_OR_Experience	54	0.202	0.143	0.361	0.007
number_breaks	54	0.260	0.058	0.299	0.028

discomfort_hips_1_0					
Variable	N	Pearson CC	Pearson P-value	Spearman CC	Spearman P-value
Age	55	0.546	<.001	0.390	0.003
Duration_since_last_day_of_exerc	55	-0.111	<.001	0.039	0.003
Frequency_of_exercising__days_we	55	0.008	<.001	0.141	0.003
Years_of_OR_Experience	54	0.299	0.028	0.386	0.004
number_breaks	54	0.212	0.123	0.343	0.011

discomfort_knees_1_0					
Variable	N	Pearson CC	Pearson P-value	Spearman CC	Spearman P-value
Age	55	0.390	0.003	0.261	0.054
Duration_since_last_day_of_exerc	55	-0.147	0.003	-0.066	0.054
Frequency_of_exercising__days_we	55	0.041	0.003	0.228	0.054
Years_of_OR_Experience	54	0.194	0.160	0.269	0.049
number_breaks	54	0.310	0.023	0.344	0.011

FIGURE 7 | Multivariate analysis.

the surgical team member in the hours preceding the operation. It would be reasonable to suggest that a provider ending an over-night shift, or a shift with a particularly heavy caseload, would experience more fatigue and body discomfort than a provider who was more rested before the operation in question. Previous studies have found that the BMI of the patient receiving the surgery could impact the level of discomfort and musculoskeletal fatigue experienced by the operating staff [12]. Other confounding factors that could influence the risk of ergonomic injuries in surgical team members include gender and hand size [13]. In fact, previous studies have shown that female surgeons were at higher risk of work-related musculoskeletal injuries than their male counterparts [14]. In addition, surgeons with smaller hand/glove size were also shown to have more surgery ergonomic related issues [15].

The findings from this study could be generalizable to other surgical fields where length of surgery in > 3 h and the surgical team is performing the surgery in the standing up position and not using any types of additional standing stools. In addition, the

relatively low cost of this intervention makes it a cost-effective way to reduce surgical team fatigue. The relative low cost also means that institution could easily adopt it as a measure to reduce workers fatigue and injuries, potentially saving on long term workers injuries related costs.

This study is limited by the lack of blinding of study participants. Blinding was not possible in this study, as the participants were aware of the presence or absence of anti-fatigue floor mats in the operating room. Future studies with interventions that are more amenable to blinding could potentially result in a lower risk of bias. In addition, the questionnaire used for this study was developed by the authors and was not a validated questionnaire. Using an ad hoc questionnaire is also a limitation. Another possible limitation to this study is the reality that not every surgery included was similar in nature. The inclusion criteria for this study required that cases were ≥ 3 h in length. This resulted in a wide variety of cases, some lasting many hours longer than others. As mentioned before, previous studies have shown that the level of discomfort and fatigue experienced by the surgical team

increases as surgery length increases. In future studies, it would be helpful to examine the cases in context to other cases of similar length, and to evaluate the effectiveness of the anti-fatigue mats at reducing discomfort and fatigue depending on different surgery lengths. However, the study was limited to surgeries being done via an open approach with the surgical team standing up for the procedure and did not include any robotic or endoscopic approaches in order to reduce confounding factors. Two of the investigators in this study also completed questionnaires during their time in the operating room, making them participants in the study; this could potentially be a source of bias. In addition, as mentioned earlier, several additional confounding factors and variables were not included in this study due to the wide variability in those factors which would have required an unrealistic large sample size to conduct the study. Examples of these factors include biometric features such as height and hand/gloves size and variables such as the type of footwear used. It is well known that surgical ergonomic health depends on multiple factors; consistent use of anti-fatigue floor mats is one of those factors. However, a number of human factors that are non-equipment related such as taking small breaks or interruption, also play an important role in ergonomic health and those are very specific to each individual. This study was limited to the use of anti-fatigue floor mats, but there is a need for more research into other ways to enhance the ergonomics of surgery, particularly as it relates to upper-body musculoskeletal fatigue.

5 | Conclusion

Floor mats are an effective and low-cost intervention to decrease the musculoskeletal symptoms and body fatigue experienced by members of the head and neck surgical team, including surgeons, assistant surgeons, and surgical scrub technicians. Regular use of floor mats in the operating room could reduce musculoskeletal injuries among surgeons, extend the number of years that surgeons are able to practice, and save hospitals billions of dollars each year.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.